AI Topics vs. AI Perception in News Media

Data Visualization Assignment Report

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Dataset Description

The visualization is based on a dataset of **news articles mentioning artificial intelligence**, enriched with metadata extracted through automated text classification. Each article includes:

- **AI sentiment**:
  - *Helper* — AI framed as beneficial
  - *Neutral* — descriptive or balanced coverage
  - *Threat* — AI framed as harmful or risky
- **AI topic category** (derived from NIST taxonomy):
- Examples include *content creation*, *cybersecurity*, *automation*, *research*, *monitoring*, *prediction*, etc.
- **Publication year**

To improve interpretability, similar categories were **grouped** (e.g., “content creation”, “content generation”, “content synthesis” → *Content*), and rare categories were combined into *Other (rare)*.

The final dataset contains several thousand articles across multiple years and dozens of AI-related topics.

`https://huggingface.co/datasets/kkChimmy/REALM`

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Visualization Type and Rationale

The main visualization is a **heatmap** showing:

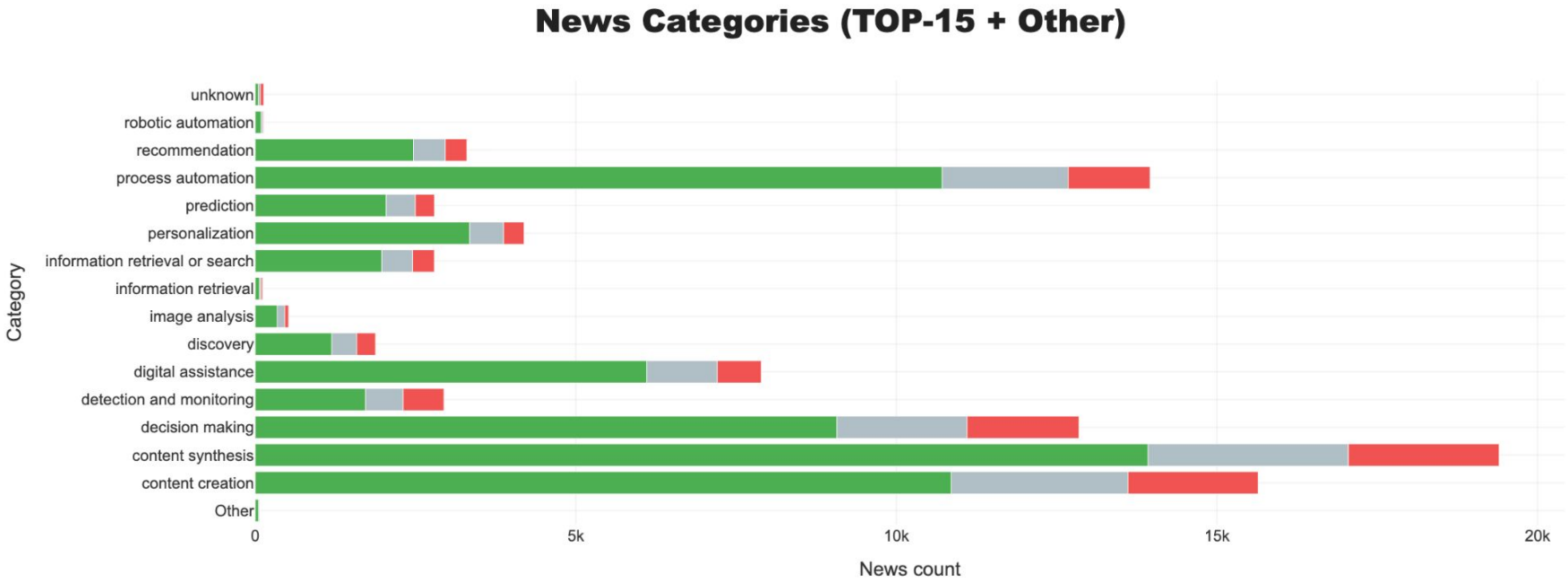
- **Rows** → grouped AI topics
- **Columns** → sentiment categories (*Helper, Neutral, Threat*)
- **Color intensity** → number of articles in each topic–sentiment combination
- **Color scale** → log-transformed to reveal structure across highly imbalanced counts

Why a heatmap?

A heatmap is ideal for this task because:

- It allows **direct comparison** across many categories simultaneously.
- It reveals **patterns of concentration** (e.g., which topics are mostly “helper”).
- It highlights **outliers** and **imbalances** that would be invisible in bar charts.
- It supports **semantic grouping** of topics, making the structure of AI coverage clearer.

Other chart types (bar charts, line charts) were used in the dashboard, but the heatmap is the most effective single visualization for conveying the core insight.


News Categories (TOP-15 + Other)
Category
unknown
robotic automation
recommendation
process automation
prediction
personalization
information retrieval or search
information retrieval
image analysis
discovery
digital assistance
detection and monitoring
decision making
content synthesis
content creation
Other
0
5k
10k
15k
20k
News count

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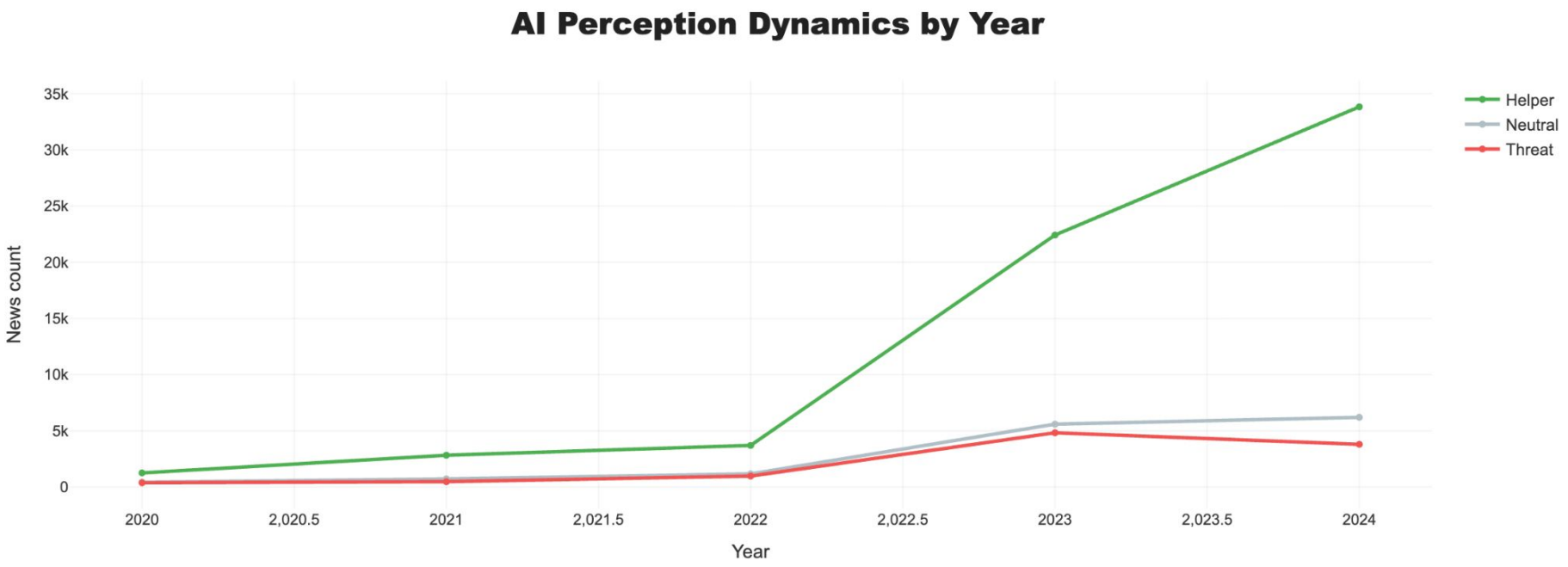
AI Perception Dynamics by Year
35k
30k
25k
20k
15k
10k
5k
0
News count
2020
2,020.5
2021
2,021.5
2022
2,022.5
2023
2,023.5
2024
Year
Helper
Neutral
Threat

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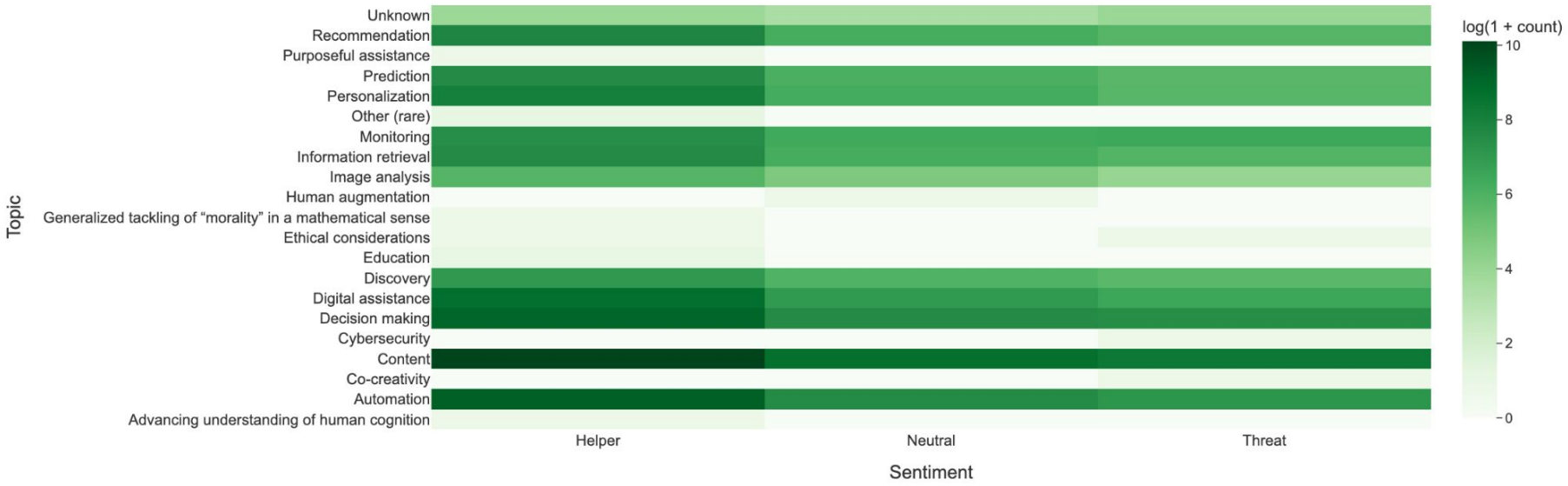
Topics vs AI Perception (log scale)



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Key Insights

1. AI is overwhelmingly framed as a “helper” across most topics.

Topics such as **Content**, **Cybersecurity**, and **Automation** show strong concentrations in the *Helper* column. This suggests that media coverage tends to emphasize AI’s usefulness and productivity benefits.

2. “Threat” framing is rare and concentrated.

Only a few topics — such as **Cybersecurity** and **Monitoring** — show noticeable “Threat” coverage. Even there, the volume is significantly lower than “Helper”.

3. Some topics are predominantly neutral.

Categories like **Ethical considerations**, **Research**, and **Education** lean toward *Neutral* sentiment, reflecting more analytical or descriptive reporting.

4. Log scale reveals structure hidden in raw counts.

Without log scaling, the heatmap would be dominated by a few very large categories. The log transform makes smaller but meaningful patterns visible.

Links

https://github.com/natharzu/Practical-Data-Science/blob/main/dataviz.ipynb

https://colab.research.google.com/gist/natharzu/da841b801fa6862805efe83e5364514d/dataviz.ipynb